

MIAE 221 · Rapid Review Sheet · Part 1

Atomic Structure, Periodic Trends & Stoichiometry

1. Materials Classes Spectrum

- **Metals:** Non-directional metallic bonds ("sea of electrons"). High ductility, high thermal/electrical conductivity, opaque, moderate-to-high stiffness.
- **Ceramics:** Ionic/covalent bonds. High hardness, extreme melting points, electrical/thermal insulators, **very brittle** (cleave along charge planes).
- **Polymers:** Covalent hydrocarbon backbones held together by weak secondary bonds. Low density, flexible, low melting points, ductile/viscoelastic.
- **Composites:** Engineered combination of matrix + reinforcement (e.g. CFRP). High strength-to-weight ratio.

2. Quantum Mechanics & Orbitals

- **Principal (n):** $n = 1, 2, 3 \dots$ Main energy level and orbital size. Max electrons per shell $= 2n^2$.
- **Azimuthal (l):** $l = 0 \dots (n - 1)$. Subshell geometry ($0 \rightarrow s, 1 \rightarrow p, 2 \rightarrow d, 3 \rightarrow f$).
- **Magnetic (m_l):** $m_l = -l \dots +l$. Spatial orientation ($2l + 1$ orbitals per subshell).
- **Spin (m_s):** $m_s = \pm 1/2$. Intrinsic electron spin direction ($\uparrow \downarrow$).
- **Aufbau Principle:** $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \dots$ ($4s$ fills before $3d$!).
- **Pauli Exclusion:** No two electrons share all 4 quantum numbers $\implies \leq 2$ electrons per orbital.
- **Hund's Rule:** Degenerate orbitals fill with parallel spins singly before pairing up.

3. Periodic Families & Electronegativity Trends

Family	Outer Config	Valence	Chemical Tendency
Inert Gas (Group 0)	$s^2 p^6$	8	Closed stable shell; chemically inert; high ionization energy.
Halogens (Group VIIA)	$s^2 p^5$	7	Needs $1 e^-$; forms -1 anions (F^-, Cl^-); high electronegativity.
Alkali Metals (Group IA)	s^1	1	Loses $1 e^-$; forms $+1$ cations (Na^+, K^+); low electronegativity.
Alkaline Earth (Group IIA)	s^2	2	Loses $2 e^-$; forms $+2$ cations (Mg^{2+}, Ca^{2+}); reactive metal.
Transition Metals	$(n-1)d^{1-10} ns^2$	Var	Incomplete inner d -orbitals; metallic bonds; multiple valences.

- **Atomic Radius:** Decreases left-to-right ($\rightarrow$), increases top-to-bottom ($\downarrow$).
- **Electronegativity (X):** Increases left-to-right ($\rightarrow$), decreases top-to-bottom ($\downarrow$). ($F = 4.0, Cs = 0.7$).

4. Quantitative Stoichiometry & Atom Count Formulas

$$\text{Moles: } n_{mol} = \frac{m}{A} = \frac{\rho \cdot V}{A} \quad \text{Atoms: } N = n_{mol} \cdot N_A = \frac{\rho \cdot V \cdot N_A}{A}$$

- $N_A = 6.022 \times 10^{23} \text{ atoms/mol} \mid 1 \text{ nm} = 10^{-7} \text{ cm} = 10^{-9} \text{ m} \mid V_{cyl} = \frac{\pi}{4} d^2 L$

5. Interatomic Forces & Energy Derivations

- **Net Energy:** $E_N(r) = E_A(r) + E_R(r) = -\frac{A}{r} + \frac{B}{r^n}$ (A : attractive, B : repulsive, $n \approx 8 - 12$).
- **Net Force:** $F_N(r) = -\frac{dE_N}{dr} = -\frac{A}{r^2} + \frac{nB}{r^{n+1}} = F_A + F_R$.
- **Equilibrium Condition (r_0):** $F_N(r_0) = 0 \iff \left. \frac{dE_N}{dr} \right|_{r_0} = 0 \implies r_0 = \left(\frac{nB}{A} \right)^{\frac{1}{n-1}}$.
- **Bonding Energy (E_0):** $E_0 = E_N(r_0) = -\frac{A}{r_0} \left(1 - \frac{1}{n} \right) = -\frac{A(n-1)}{nr_0}$.